

A Formally Verified Divisibility Theorem for A211420

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Ordinary proof of the explicit L_r^r strengthening generated with OpenAI GPT-5.6 Pro.
Primary Lean 4 formalization and repository preparation by OpenAI Codex; formal proof checked by Lean's kernel.
A separate second Lean formalization was generated by Aristotle (Harmonic).
Initiated, curated, and published by Danny Cabezas.

1 Statement

For $n, r, k \in \mathbb{N}$, put

$$A_{\text{num}}(n) = (8n)!n!, \quad A_{\text{den}}(n) = (4n)!(3n)!(2n)!,$$
$$D_{k,r}(n) = \prod_{i=1}^r (kn + i), \quad L_r = \text{lcm}(1, 2, \dots, 8r - 1).$$

Theorem 1 (Strengthened A211420 divisibility). *For every $n \geq 0$, every $r \geq 1$, and every $k \in \{1, 2, 3\}$,*

$$A_{\text{den}}(n) \mid A_{\text{num}}(n)$$

and

$$D_{k,r}(n)A_{\text{den}}(n) \mid L_r^r A_{\text{num}}(n).$$

Consequently, with the exact integer quotient

$$a_n = \frac{A_{\text{num}}(n)}{A_{\text{den}}(n)},$$

one has $L_r^r > 0$ and

$$D_{k,r}(n) \mid L_r^r a_n.$$

2 Prior work and contribution boundary

The integrality of a_n was already known from Bober's classification: it is the $a = 4$, $b = 1$ specialization of his infinite family of integral factorial ratios [1, Theorem 1.2, family (8)]. The present Lean development formalizes this integrality rather than assuming it.

The weaker assertion that, for each fixed $k \in \{1, 2, 3\}$ and $r \geq 1$, some positive constant $C(k, r)$ exists follows from Landau's multivariate factorial-ratio criterion after a two-variable specialization; for example,

$$C(k, r) = ((8r)!)^r$$

is sufficient. This is background identified in the accompanying audits and is not the formally verified contribution claimed here. The result proved and formalized here is the stronger explicit bound

$$C(k, r) = L_r^r.$$

No claim of minimality or historical priority is made. The searches conducted found no prior source for this full explicit strengthening.

Two earlier results give exact or closely related special cases. Sun's Theorem 1.3(ii), specialized with his parameter $k = 4$, yields

$$(3n + 1) \mid 5a_n$$

for every n [3]. Yang's Theorem 1, specialized with $a = 4$ and $b = 1$, yields

$$(2n + 1)(2n + 3) \mid 99a_n$$

for every positive n [4]. Yang's second factor is $2n + 3$, so this is related to the earlier odd-progression problem rather than the consecutive $r = 2$ product in Theorem 1.

3 Ordinary mathematical proof

For a prime p , initially interpret v_p as the valuation on positive rational numbers. Define the one-periodic Landau step function

$$\Delta(x) = \lfloor 8x \rfloor + \lfloor x \rfloor - \lfloor 4x \rfloor - \lfloor 3x \rfloor - \lfloor 2x \rfloor.$$

The coefficient balance $8 + 1 - 4 - 3 - 2 = 0$ gives $\Delta(x + 1) = \Delta(x)$. On $[0, 1)$ the relevant breakpoints are

$$0, \frac{1}{8}, \frac{1}{4}, \frac{1}{3}, \frac{3}{8}, \frac{1}{2}, \frac{5}{8}, \frac{2}{3}, \frac{3}{4}, \frac{7}{8}, 1.$$

Direct evaluation on the intervening half-open intervals gives

$$\Delta(x) = 1 \iff \{x\} \in \left[\frac{1}{8}, \frac{1}{3}\right) \cup \left[\frac{3}{8}, \frac{1}{2}\right) \cup \left[\frac{5}{8}, \frac{2}{3}\right) \cup \left[\frac{7}{8}, 1\right),$$

and $\Delta(x) = 0$ otherwise. In particular, $\Delta \geq 0$.

Legendre's formula gives the finite identity

$$v_p\left(\frac{A_{\text{num}}(n)}{A_{\text{den}}(n)}\right) = \sum_{j \geq 1} \Delta\left(\frac{n}{p^j}\right).$$

Indeed, all summands vanish when $p^j > 8n$. Every prime valuation of the positive rational factorial quotient is therefore nonnegative. It follows that $A_{\text{den}}(n) \mid A_{\text{num}}(n)$ and a_n is a positive integer.

For a prime power $q = p^j$, define

$$N_q = \#\{i \in \{1, \dots, r\} : q \mid kn + i\}.$$

Then

$$v_p(D_{k,r}(n)) = \sum_{j \geq 1} N_{p^j},$$

again with only finitely many nonzero terms.

Suppose first that $q \geq 8r$. Then $q > r$, so at most one integer among $kn + 1, \dots, kn + r$ is divisible by q and $N_q \leq 1$. If $q \mid kn + i$, write $kn + i = mq$. Then

$$\frac{n}{q} = \frac{m}{k} - \frac{i}{kq}, \quad 0 < \frac{i}{kq} \leq \frac{1}{8k}.$$

For $k = 1$, the fractional part of n/q lies in $[7/8, 1)$. For $k = 2$, according as m is even or odd, it lies in $[15/16, 1)$ or $[7/16, 1/2)$. For $k = 3$, according to m modulo 3, it lies in

$$[23/24, 1), \quad [7/24, 1/3), \quad [5/8, 2/3).$$

Every displayed interval is contained in the support of Δ . The included left endpoints cover the equality case $q = 8r$ and $i = r$. Consequently

$$N_q \leq \Delta(n/q) \quad (q \geq 8r).$$

For $q = p^j < 8r$, use the elementary bound $N_q \leq r$. The exact definition of L_r gives

$$v_p(L_r) = \#\{j \geq 1 : p^j < 8r\}.$$

Therefore, for every prime p ,

$$\begin{aligned} v_p(D_{k,r}(n)) &= \sum_{p^j < 8r} N_{p^j} + \sum_{p^j \geq 8r} N_{p^j} \\ &\leq r v_p(L_r) + \sum_{p^j \geq 8r} \Delta(n/p^j) \\ &\leq r v_p(L_r) + \sum_{j \geq 1} \Delta(n/p^j) \\ &= v_p(L_r^r a_n). \end{aligned}$$

The second inequality uses $\Delta \geq 0$ for the omitted small powers. Prime-by-prime comparison proves $D_{k,r}(n) \mid L_r^r a_n$. Finally, L_r is a nonzero least common multiple, so $L_r^r > 0$. This proves Theorem 1.

4 Lean verification

The Lean project defines

$$\text{L r} = \text{Nat.lcmUpto } (8 * \text{r} - 1),$$

which is exactly the lcm of $1, \dots, 8r - 1$. The principal declarations are

```
A211420.integral_cross
A211420.strengthened_cross
A211420.constant_pos
A211420.ADen_mul_a
A211420.strengthened
```

The primary theorem is division-free; the quotient corollary is derived only after exact factorial-ratio integrality is established.

The project is pinned to Lean 4.30.0, commit d024af099ca4bf2c86f649261ebf59565dc8c622, and mathlib v4.30.0, commit c5ea00351c28e24afc9f0f84379aa41082b1188f. From the repository root, run

```
./scripts/verify.sh
```

to build the project and reject forbidden proof escapes in project Lean files.

5 Scope and provenance boundary

The theorem displayed above is kernel-checked. Finite computations were used only for falsification and diagnosis, not as proof. Four AI-generated adversarial audit records distributed with the repository are not human peer review. No claim of historical priority, minimality of L_r^r , or formal verification of additional proposed variants is made.

The searches supplied with this project found no prior source for the exact L_r^r strengthening, but bounded searches cannot certify originality. The integrality background belongs to the classical Landau factorial-ratio framework; see Bober’s classification and the work of Rodriguez-Villegas.

References

- [1] J. W. Bober, *Factorial ratios, hypergeometric series, and a family of step functions*, J. London Math. Soc. (2) **79** (2009), 422–444.
- [2] E. Landau, *Sur les conditions de divisibilité d’un produit de factorielles par un autre*, Nouvelles annales de mathématiques, 3e série, **19** (1900), 344–362.
- [3] Z.-W. Sun, *On divisibility of binomial coefficients*, J. Aust. Math. Soc. **93** (2012), 189–201, doi:10.1017/S1446788712000171.
- [4] Q.-H. Yang, *Proof of a conjecture of Amdeberhan and Moll on a divisibility property of binomial coefficients*, Electron. J. Combin. **22** (2015), Paper P1.9; arXiv:1401.1108.
- [5] F. Rodriguez-Villegas, *Integral ratios of factorials and algebraic hypergeometric functions*, arXiv:math/0701362 (2007).